

Clonidine as Monotherapy for Neonatal Opioid Withdrawal Syndrome: A Randomized Trial

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abstract

OBJECTIVE: We sought to determine whether clonidine, a non-opioid α -2-adrenergic agonist, would effectively treat neonatal opioid withdrawal syndrome (NOWS).

METHODS: This was an intention-to-treat randomized clinical trial. Enrollment criteria included prenatal opioid exposure, age ≤ 7 days, gestational age ≥ 35 weeks, no other medical condition, and need for pharmacotherapy. Primary outcomes were length of treatment and neurobehavioral performance.

RESULTS: A total of 1107 patients were screened for enrollment (645 ineligible, 91 parents or staff unavailable, 216 declined, 155 consented). Of 155 infants, 120 required treatment and were randomized to receive oral clonidine ($n = 60$) at 1 $\mu\text{g/kg/dose}$ or morphine ($n = 60$), 0.06 mg/kg/dose, every 3 hours. Infants with no improvement had their doses increased by 25% of the initial dose every 12 to 24 hours. Those without improvement by the fourth dose increase, received adjunct therapy. Length of treatment did not differ between morphine and clonidine, with median (95% confidence interval [CI]) days, respectively, of 15 (13–17) and 17 (15–19), $P = .48$. More clonidine-treated infants (45%) needed adjunct therapy versus 10% in the morphine group, adjusted odds ratio (95% CI) = 8.85 (2.87–27.31). After treatment completion, the NICU Network Neurobehavioral Scales summary scores did not differ between clonidine-treated and morphine-treated infants.

CONCLUSIONS: Length of pharmacologic treatment and final neurobehavioral performance were not significantly different between the clonidine- and morphine-treated groups. Clonidine appears to be an effective non-opioid medication to treat NOWS. Future studies are needed to determine the optimal clonidine dosage for a quicker response and obviation of adjunct therapy.



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This trial has been registered at www.clinicaltrials.gov (identifier NCT03396588).

Deidentified individual participant data will not be made available for sharing based on participant institutional review board consent.

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WHAT'S KNOWN ON THIS SUBJECT: Morphine is a commonly used treatment of neonatal opioid withdrawal syndrome (NOWS) despite concerns about its adverse effects on the developing brain. Clonidine, a non-opiate α -2 adrenergic agonist, is reported to be effective as an adjunct therapy in NOWS treatment.

WHAT THIS STUDY ADDS: Clonidine, as a non-opiate single-drug treatment for NOWS, is not significantly different from morphine regarding the duration of treatment and infant neurobehavioral outcomes post-treatment. Clonidine appears to be an effective non-opioid medication for NOWS therapy.

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The opioid epidemic in the United States has resulted in increasing overdose deaths,¹ maternal death,² and neonatal abstinence syndrome (NAS),³ which is associated with short-term and long-term consequences.^{4,5} From 2010 to 2017, NAS rates increased from 4.0/1000 to 7.3/1000 hospital births; rates varied across states from 1.3/1000 to 53.5/1000 births by region,^{3,6} urbanicity,⁷ or county.^{8,9} These increased NAS rates were related to prenatal opioid exposure with or without other substances, lately referred to as neonatal opioid withdrawal syndrome (NOWS). In utero exposure occurs from opioid misuse or as prescribed for chronic pain relief or medication for opioid use disorder (MOUD). NOWS has been associated with increased health care costs^{4,10,11} from the prolonged length of hospital stay (LOS) related to the length of treatment (LOT).¹⁰

The reported adverse effects of in utero opioid exposure on the developing brain include hydrocephaly,^{12,13} neonatal stroke,¹⁴ small head size,^{15–23} decreased brain cell number,²⁴ postnatal head growth deceleration,^{25,26} reduced brain volumes,²⁷ and associated long-term cognitive deficits and behavioral problems.^{28,29} Despite the detrimental effects of in utero opioid exposure, the opiate morphine remains a common first-line drug for NOWS treatment,^{30–32} resulting in additional weeks to months of exposing the developing brain to opioids.^{33–35}

Prolonged opioid exposure results in the activation of opioid receptors in the locus coeruleus, in which clusters of noradrenergic cells are collocated.^{36,37} Chronic opioid supply inhibits noradrenergic cells; the discontinuation of opioid exposure increases noradrenergic activity.³⁸ Clonidine inhibits noradrenaline release, resulting in decreased noradrenergic activity and opioid withdrawal manifestations.^{39,40} Clonidine is a reported effective single-drug therapy in a small number of babies with NOWS^{41–43} and as an adjunct therapy.^{44–46} In preclinical studies, clonidine has advantages over morphine; it stabilizes breathing and temperature control that is disrupted by morphine,⁴⁷ decreases ketamine-induced apoptosis and brain injury in hypoxic-ischemic brain damage,⁴⁸ suppresses cytokines, and downregulates opioid receptor expression in neonatal mononuclear cells.⁴⁹

In our previous study, clonidine, compared with morphine for NAS treatment, was associated with shorter LOT and improved neurobehavioral performance.⁵⁰ This supported evaluating a non-opioid drug as an alternative to treat NOWS. We sought to determine if the non-opioid α -2 adrenergic agonist, clonidine, would effectively treat NOWS.

METHODS

This study (<https://clinicaltrials.gov/ct2/show/NCT03396588>) had approval from the institutional review board (IRB) and a certificate of confidentiality from the National Institute on Drug Abuse. Parent(s) gave informed consent for study participation. This study was an intention-to-treat, double-blind trial, with treatment randomized between clonidine

and morphine, carried out at the Kentucky Children's Hospital. Primary outcomes included LOT and post-treatment neurobehavioral status, which is a reported predictor of childhood outcomes.⁵¹

Research nurses screened all admissions for prenatal substance exposure at the well-baby nursery and the NICU. Enrollment criteria included (1) gestational age ≥ 35 weeks, (2) any prenatal opioid exposure with or without other substances from the mother reporting opioid use or positive toxicology results, (3) ≤ 7 days postnatal age, (4) not unlikely to survive, (5) no medical condition other than NOWS, (6) no seizures, (7) no prenatal cocaine exposure, (8) meets treatment criteria for NOWS, and (9) informed consent. Prenatal cocaine exposure was excluded because tone abnormalities have been attributed to cocaine central nervous system effects rather than drug withdrawal.^{52,53}

All infants had non-pharmacological intervention (swaddling, low noise, and lighting environment, rooming in, infant massage, etc).^{54–56} Throughout the hospital stay, infants were monitored using the Finnegan Neonatal Abstinence Scoring System.⁵⁵ The Finnegan Neonatal Abstinence Scoring System scores (FS) guided the initiation and monitoring of treatment, allowing for comparison with other treatment studies.^{45,57,58} Clonidine or morphine was started when an infant had 3 consecutive FSs ≥ 8 or 2 consecutive FSs ≥ 12 with the evaluation and approval of the attending physician. Treatment assignment was obtained through the Investigational Drug Service of the Center for Clinical and Translational Science. Clinical pharmacists prepared and dispensed the study drug, with clonidine and morphine having identical appearances. Block randomization allowed for balanced enrollment in each treatment arm. Families, providers, and research staff were masked to the treatment assignment. Our initial drug dosage (clonidine, 1 $\mu\text{g/kg/dose}$ or morphine, 60 $\mu\text{g/kg/dose}$) approximated our previous study's receiver operating characteristics that provided the best control of signs⁵⁰ and was given every 3 hours concomitant with the infant's feeding schedule. The initial clonidine dose was within the range given in other studies^{42–46} and supported by published population pharmacokinetics.⁵⁹

The dose for either drug was increased by 25% of the initial dose every 12 to 24 hours until FSs were consistently < 8 , to a maximum of 2 $\mu\text{g/kg/dose}$ for clonidine and 120 $\mu\text{g/kg/dose}$ for morphine. The maximum clonidine dose of 16 $\mu\text{g/kg/day}$ was slightly higher than reported by Agthe et al⁴⁵ but within the dose range reported by O'Mara et al.⁶⁰

After 24 to 48 hours of stabilization, dose weaning began by 10% of the maximum dose every 24 hours. When FSs increased during weaning, the dose was increased to the previous dose; weaning resumed after 24 to 48 hours. If no improvement was noted at the maximum dose, phenobarbital was added at a loading dose of 10 mg/kg and

maintenance dose of 2.5 mg/kg q 12 hours.⁶¹ Some attending physicians preferred clonidine or morphine as an adjunct. Thus, infants were allowed to receive either morphine or clonidine as the secondary drug, which the infant was not randomly assigned to receive. Masking was maintained. After stabilization after the addition of the adjunct drug, the primary (first) drug was weaned daily by 10% of the maximum dose. After discontinuation of the primary drug, adjunct drug weaning followed. After 48 hours of observation without medication, infants were discharged. Infants on phenobarbital were discharged with a weaning schedule over 3 weeks.

Data Collection

Drug use during pregnancy was obtained through maternal interviews by research staff, who also abstracted medical records for demographic information, including race and ethnicity, maternal pregnancy history, drug use, MOUD, delivery information, FSs, dose changes, duration of treatment, and drug effects, conditions, complications for reporting to the IRB and Drug Safety Monitoring Board (DSMB) of the Center for Clinical and Translational Science. Data entry and management were conducted by using Research Electronic Data Capture tools (REDCap 11.4.2 - 2021 Vanderbilt University).^{62,63}

Neurobehavior Assessment

The NICU Network Neurobehavioral Scale (NNNS)^{64,65} requires eliciting responses from a sleep or quiet state to crying (behavioral state 6). Infants with severe withdrawal are typically unable to complete the assessment. Therefore, after withdrawal signs improved and dose weaning began, trained and certified research nurses, who were blinded to treatment assignment, performed the initial assessment; this was repeated at treatment completion. The NNNS assesses neurologic, behavioral, and stress/abstinence components from the Brazelton Scale⁶⁶ and Finnegan's work,⁶⁷ yielding 13 summary scores:⁶⁵ habituation, attention, handling, quality of movement, regulation, non-optimal reflexes, asymmetry total/reflexes, stress/abstinence, arousal, hypertonicity, hypotonicity, excitability, and lethargy.

Statistical Methods

When designing this study, we anticipated recruiting ~200 infants. Assuming 20% attrition (80 infants in each treatment group), there would be at least 80% power to detect a difference with an effect size of at least 0.45 when using a 2-sided test with a 5% significance level. This effect size was exceeded for some outcomes in the pilot data.⁵⁰ An interim analysis was conducted in 2020 for the DSMB. At that time, 94 subjects had been randomly assigned. The study was allowed to continue.

Based on Consolidated Standards of Reporting Trials (CONSORT) standards, baseline demographic variables were summarized for morphine and clonidine treatment.^{68,69}

Frequencies (%) are presented for categorical variables and either mean (SD) or median (min, max) for continuous variables. For outcome results, we present 95% confidence intervals (CIs).

A log-rank test was used to compare treatment arms regarding LOT and LOS.⁷⁰ To use data from infants withdrawn from the study, the time of withdrawal was specified as the censoring time. Cox proportional hazards regression modeling was used in a follow-up analysis using treatment and selected risk factors associated with NOWS severity (birth weight, gestational age, gender, cesarean delivery, maternal psychiatric disorder, inborn status, MOUD, other opioid, and non-opioid substance use).^{32,51,71–74} Because smoking was highly prevalent and alcohol use was infrequent, they were not included in the model. In post hoc analyses, a negative binomial regression was used to model LOT for 115 infants, excluding those withdrawn from the study. Empirical SE estimates were employed to ensure valid inference.

Treatment arms were compared regarding the need for adjunct medication using frequencies (%) and χ^2 tests. Of 5 infants withdrawn, only 1 received adjunct medication; the other 4 were observed for <2 days. Utilizing data from all 120 infants, we assumed that the 4 infants did not require adjunct treatment. We also removed these 4 infants in a follow-up analysis. The remaining 116 infants were used to fit a logistic regression model, from which adjusted odds ratios were derived.

Wilcoxon rank tests were used to compare treatment groups regarding the timing between initial and final NNNS assessments and the NNNS scores (initial, final, and change). *P* values were adjusted using the false discovery rate method because of multiple comparisons across 13 summary scores.⁷⁵

Intention-to-treat analyses used all available data based on treatment assignment. All tests were 2-sided, with a statistical significance of *P* < .05. Analyses were conducted in SAS v9.4 (SAS Institute Inc 2013. SAS/ACCESS® 9.4 Interface to ADABAS: Reference. Cary, NC: SAS Institute Inc.).

RESULTS

The enrollment period started in December 2017 and ended in February 2022, at funding completion. In 2020 and 2021, coronavirus disease 2019 restrictions over face-to-face encounters with subjects resulted in decreased enrollment. Enrollment continued through phone consent and follow-up post-discharge through virtual visits.

The CONSORT diagram is in Fig 1. Infants with prenatal substance exposure (*n* = 1107) were assessed for eligibility. There were 155 whose parents gave consent, but 35 did not meet the treatment criteria. The remaining 120 infants were randomly assigned to receive clonidine (*n* = 60) or morphine (*n* = 60). Five infants were withdrawn from the study (clonidine [*n* = 2] and morphine

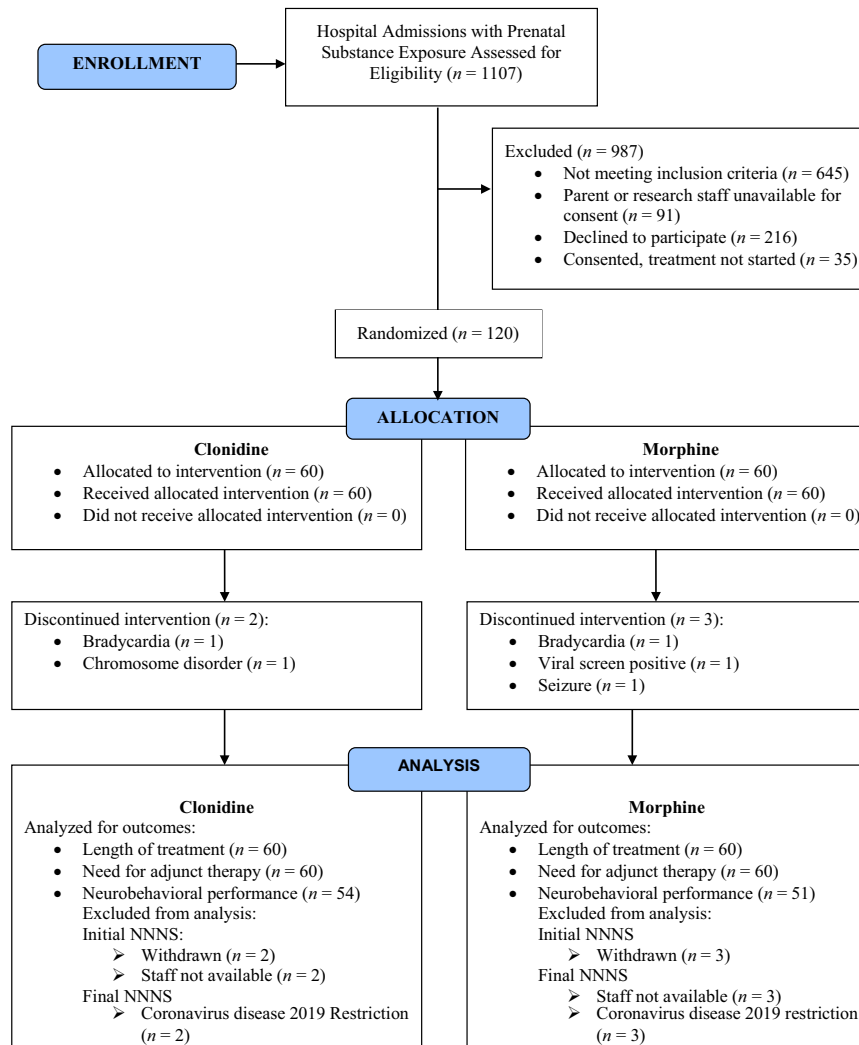


FIGURE 1

CONSORT flow diagram showing the number of participants screened for eligibility who were consented for enrollment, enrolled, received allocation, were excluded or withdrawn from participation after the intervention initiated and the reasons, and included in the outcome analysis.

[$n = 3$]). Four of 5 infants were withdrawn after 1 to 2 days of treatment; 1 randomly assigned to receive clonidine was withdrawn after 10 days of treatment because of abnormal chromosome results.

Table 1 reveals characteristics of mothers and infants by treatment with no post hoc comparison, per the CONSORT statement.^{68,69} A total of 52 (87%) and 49 (82%) mothers in the clonidine and morphine groups, respectively, were receiving MOUD. Despite this, the use of other substances was prevalent (Table 2) based on maternal self-report and toxicology results.

Before treatment, the median (95% CI) of the maximum FS was similar in clonidine- and morphine-treated infants [14.5 [14.1–15.6] and 14 [13.9–15.5], respectively). FS decreased subsequently. By the third day of treatment, the maximum FS was significantly lower ($P < .001$) in the morphine-treated group, with a median

(95% CI) of 8.0 (7.3–8.6) versus 11.5 (11.0–12.5) in the clonidine group. In Fig 2A–B, there are similar patterns of mean FS (95% CI) over time for the treatment groups and subgroups with monotherapy. Mean FS scores were similar at treatment initiation, with a more pronounced drop by the second day of treatment with morphine treatment. After 6 days, the mean daily scores of the clonidine group trended lower than the morphine group, although the CIs overlapped.

Duration of Treatment

The LOT did not differ significantly between treatments ($P = .48$); the medians (95% CI) were 17 (15–19) and 15 (13–17) days, respectively, for the clonidine and morphine groups. Neither was there a significant difference in LOS between clonidine and morphine groups, with medians (95% CI) of 22 (20–24) and 19 (17–21) days,

TABLE 1 Characteristics of Mothers and Infants Enrolled in the Study by Treatment Assignment							
	Clonidine (n = 60)	Morphine (n = 60)	All (n = 120)		Clonidine (n = 60)	Morphine (n = 60)	All (n = 120)
Maternal Characteristics	n (%)	n (%)	n (%)	Infant Characteristics	n (%)	n (%)	n (%)
Age, y, mean (SD)	29.3 (4.8)	29.5 (4.3)	29.4 (4.5)	Gest age, wk, mean (SD)	38.7 (1.2)	38.3 (1.3)	38.5 (1.3)
Race and ethnicity ^a				Race and ethnicity ^a			
				Non-Hispanic white	53 (88)	50 (83)	103 (86)
				Non-Hispanic Black	0 (0)	0 (0)	0 (0)
Non-Hispanic white	57 (95)	59 (98)	116 (97)	Non-Hispanic multiracial	6 (10)	6 (10)	12 (10)
Non-Hispanic Black	1 (2)	0 (0)	1 (1)	Non-Hispanic American Indian	1 (2)	0 (0)	1 (1)
Non-Hispanic multiracial	1 (2)	1 (2)	2 (2)	Hispanic white	0 (0)	2 (3)	2 (2)
Non-Hispanic American Indian	1 (2)	0 (0)	1 (1)	Hispanic multiracial	0 (0)	2 (3)	2 (2)
Insurance				Preterm <37 wk	3 (5)	4 (7)	7 (6)
Medicaid	58 (97)	59 (98)	117 (98)				
Private	2 (3)	1 (2)	3 (2)				
Rural	42 (70)	46 (77)	88 (73)	Males	37 (62)	30 (50)	67 (56)
Limited or no prenatal care	23 (38)	24 (40)	47 (39)	Inborn ^b	27 (45)	27 (45)	54 (45)
Hospitalization, this pregnancy before delivery	9 (15)	12 (20)	21 (18)	Birth weight, mean (SD) g	3144 (497)	3037 (496)	3091 (497)
Gravida, median (min, max)	4 (1,10)	3 (1,10)	4 (1, 10)	Birth head circ, mean (SD) cm	33.9 (1.7)	33.6 (1.6)	33.7 (1.6)
Para, median (min, max)	3 (1, 7)	3 (1, 9)	3 (1, 9)	Birth length, mean (SD) cm	48.1 (2.4)	48.3 (2.1)	48.2 (2.2)
Hypertension	9 (15)	7 (12)	16 (13)	1 min Apgar median (min, max)	8 (1,9)	8 (4,9)	8 (1,9)
Diabetes	3 (5)	6 (10)	9 (8)	5 min Apgar median (min, max)	9 (5,10)	9 (7, 10)	9 (5, 10)
Hepatitis C positive	26 (43)	34 (57)	60 (50)	Intrauterine growth restriction	6 (10)	7 (12)	13 (11)
Hypothyroid	3 (5)	10 (17)	13 (11)	Cesarean section birth	31 (52)	29 (48)	60 (50)
Psych/emotional illness	23 (38)	19 (32)	42 (35)	Not breastfed	40 (67)	35 (58)	75 (63)
MOUD	52 (87)	49 (82)	101 (84)	Post-natal age at treatment start, mean (SD) d	2.7 (1.3)	2.5 (1.2)	2.6 (1.3)
Buprenorphine	26 (50)	30 (61)	56 (55)				
Buprenorphine/naloxone	21 (40)	16 (33)	37 (37)				
Methadone	5 (10)	3 (6)	8 (8)				
Tobacco	51 (85)	52 (87)	103 (86)				
Alcohol	0 (0)	3 (5)	3 (3)				

^a Race and ethnicity were derived from review of electronic medical records.

^b Birth at the hospital study site versus those transferred in to study site from another birth facility.

respectively ($P = .30$). In those receiving a single therapy, the LOT was similar for clonidine and morphine subgroups, with medians (95% CI) of 14 (12–16) days and 14 (12–17) days, respectively ($P = .59$). The mean (SD) total morphine dose per patient in the clonidine arm was 5.5 (13.2) mg and 9.1 (19.8) mg in the morphine arm. The clonidine-treated infants who also received morphine had a mean (SD) of 27.7 (16.0) mg total morphine dose per patient; the morphine treatment duration was longer, starting a few days from clonidine initiation and weaning beginning only after clonidine discontinuation.

In an unadjusted model, the hazard ratio (HR) for clonidine to morphine was estimated to be 0.88 (95% CI: [0.61–1.28], $P = .50$). In the adjusted model (Table 3), the results were similar, (HR = 0.82, 95% CI: [0.55–1.23], $P = .33$). A heavier birth weight was associated with longer LOT (HR = 0.95, 95% CI: [0.91–0.99], $P = .01$). Exposure to any other opioids predicted a shorter LOT, but not significantly when exposures were only based on toxicology results.

When excluding infants who were withdrawn from the study, the results were similar (Supplemental Table 5). The unadjusted negative binomial model showed an estimated mean duration of treatment of clonidine to be 18.8 days and 18.0 days for morphine, corresponding to a ratio of 1.05 (95% CI: 0.87–1.26).

Need for Adjunct Medication

More clonidine-treated infants required adjunct therapy than the morphine-treated infants, as follows: 27/60 (45%) and 6/60 (10%), respectively ($P < .001$). The results were similar when excluding the withdrawn subjects ($P < .001$). The adjusted odds for adjunct therapy were higher for clonidine (odds ratio = 8.85, 95% CI: [2.87–27.31], $P < .001$; Table 3). No other factor except heavier birth weight was associated with adjunct therapy (odds ratio = 1.17, 95% CI [1.04–1.31]).

Neurobehavioral Outcome Measures

For the clonidine group ($n = 56$), the median (min, max) interval between treatment initiation and the initial

TABLE 2 Exposures to Other Substances (Opioid and Non-Opioids) by Treatment Groups From Maternal Self-Report or Toxicology Results Compared With Exposures Based Only on Toxicology Results

	Clonidine (<i>n</i> = 60)		Morphine (<i>n</i> = 60)	
	Maternal Report or Toxicology, <i>n</i> (%)	Toxicology Only, <i>n</i> (%)	Maternal Report or Toxicology, <i>n</i> (%)	Toxicology Only, <i>n</i> (%)
Opioids				
Codeine	3 (5)	2 (3)	3 (5)	2 (3)
Hydrocodone	8 (13)	1 (2)	5 (8)	0 (0)
Hydromorphone	5 (8)	4 (7)	2 (3)	0 (0)
Morphine	8 (13)	4 (7)	6 (10)	6 (10)
Heroin	14 (23)	3 (5)	10 (17)	1 (2)
Oxycodone	9 (15)	3 (5)	11 (18)	6 (10)
Methadone*	3 (5)*	2 (3)*	1 (2)*	1 (2)*
Fentanyl	14 (23)	7 (12)	12 (20)	7 (12)
Buprenorphine*	5 (8)*	4 (7)*	7 (12)*	3 (3)*
Tramadol	3 (5)	2 (3)	3 (5)	3 (5)
Non-opioids				
Tobacco**	51 (85)	**	52 (87)	**
Alcohol	0 (0)	0 (0)	3 (5)	1 (2)
Amphetamines/ methamphetamine	19 (32)	11 (18)	15 (25)	10 (17)
Benzodiazepines	8 (13)	5 (8)	5 (8)	2 (3)
Cannabinoids	23 (38)	14 (23)	9 (15)	3 (5)
Gabapentin**	12 (20)	**	12 (20)	**
Phencyclidine	0 (0)	0 (0)	0 (0)	0 (0)
Barbiturates	1 (2)	1 (2)	2 (3)	1 (2)
SSRI (selective serotonin reuptake inhibitor)**	10 (17)	**	10 (17)	**

* Only included are exposures other than MOUD.

** Toxicology test not performed.

NNNS examination was 5 (1–16) days (1 outlier), which is significantly longer compared with 3 (1–9) days in the morphine-treated (*n* = 57) group ($P < .001$). For a post hoc comparison of subgroups, clonidine only (*n* = 31) and morphine only (*n* = 51), the median (min, max) intervals, respectively, were 3 (1–8) days and 3 (1–9) days ($P = .52$).

The interval between treatment initiation and the final NNNs examination did not differ significantly between treatment groups; for clonidine (*n* = 54), the median (min, max) was 16.5 (8–48) days, and for morphine (*n* = 51), it was 15 (7–50) days ($P = .24$). For monotherapy subgroups, clonidine (*n* = 32) and morphine (*n* = 45), the intervals were 14 (8–24) and 14 (7–32), respectively, and $P = .94$.

The initial, final, and change in the NNNs summary scores of the treatment groups are shown in Table 4. The results are similar when comparing monotherapy subgroups (Supplemental Table 6). At the initial testing, clonidine-treated babies scored worse on arousal, hyper-tonicity, and stress abstinence. However, between the initial and final assessments, with the duration being similar between groups, the clonidine-treated infants showed significant improvement (less handling, excitability, stress abstinence, and improved arousal and regulation); thus, after treatment completion, clonidine-treated infants did as well as those who were morphine-treated regarding neurobehavioral performance.

Monitoring of Adverse Effects

Conditions monitored and reported to both the DSMB and IRB are in Supplemental Table 7. These conditions were rare and may not be related to treatment.

DISCUSSION

In this randomized trial comparing the clonidine and morphine treatment of NOWS, we found no significant differences in LOT and LOS. However, nearly one-half of the infants treated with clonidine received adjunct therapy. The FS decreased with the initiation of treatment, but the decrease in scores in clonidine-treated babies lagged behind that of morphine-treated infants by 1 to 2 days, resulting in the initiation of adjunct therapy. Initially, clonidine-treated infants had worse neurobehavioral performance scores but were no longer different from morphine-treated infants at treatment completion within similar intervals between the initial and final assessments.

The lack of difference in LOT between groups is notable, considering a larger proportion of clonidine-treated babies receiving adjunct therapy and adjustment for multiple factors. We were able to successfully provide an opioid-sparing medication-assisted treatment regimen to a cohort of infants with significant NOWS. This use of non-opioids may be extended for conditions other than

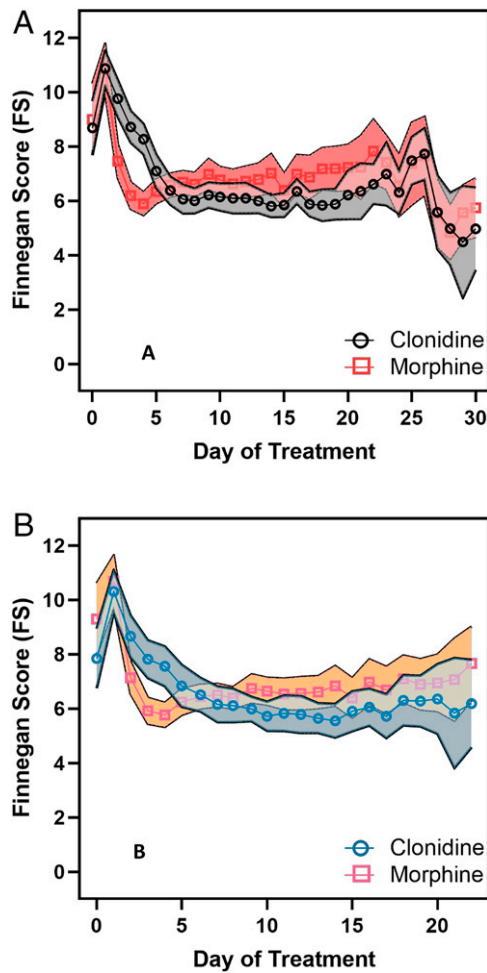


FIGURE 2

Trajectories of the Finnegan scores, mean, and 95% CI over the course of treatment of all infants in the treatment groups (Fig 2A) and for those who only received a single drug therapy and no adjunct medication (Fig 2B). The means and 95% CIs are descriptive and conditional on subjects still being treated on the given day.

NOWS, such as for the relief of chronic pain or in postoperative states.

A high proportion of clonidine-treated infants needing adjunct therapy is a concern. However, this could be addressed by adjusting the dosing schedule. On the basis of our pilot trial results of conservatively starting clonidine dosing at 0.5 µg/kg,⁵⁰ published data,⁷⁶ and a plan to monitor possible adverse events, we started clonidine at 1 µg/kg. Considering that the estimated half-life of clonidine in a neonate is 14 hours,⁵⁹ compared with 6.5 hours^{77,78} for morphine, it is reasonable to assume that the maximum pharmacologic effects would be near the time of achieving steady-state, which is typically 5-fold the agent's half-life, or ~70 hours for clonidine and ~32 hours for morphine. This is consistent with the lag in the decrease of FS by 1 to 2 days with clonidine treatment. Conversely, the subsequent trend

TABLE 3 Adjusted Results From Cox Proportional Hazards Regression Model and Logistic Regression Model Based on Exposures

Adjusted Results Predicting LOT Cox Proportional Hazards Regression Model				Adjusted Results Predicting Adjunct Therapy Logistic Regression Model				
Variables	Substance Exposure All Sources ^a		Substance Exposure Toxicology ^b		P	Substance Exposure All Sources ^a	Substance Exposure Toxicology ^b	
	HR (95% CI)	P	HR (95% CI)	P			Odds Ratio (95% CI)	Odds Ratio (95% CI)
Treatment group								
Clonidine	0.82 (0.55–1.23)	.33	0.90 (0.61–1.32)	.58	<.001	8.85 (2.87–27.31)	8.22 (2.72–24.91)	
Morphine	Reference		Reference			Reference	Reference	
Gestational age, wk	0.92 (0.75–1.12)	.41	0.97 (0.79–1.18)	.75	.46	1.20 (0.74–1.96)	1.21 (0.74–1.99)	
Birth Weight, 100 g	0.95 (0.91–0.99)	.01	0.95 (0.91–0.99)	.01	.007	1.17 (1.04–1.31)	1.17 (1.04–1.31)	
Sex								
Female	1.04 (0.70–1.56)	.85	1.01 (0.67–1.54)	.95	.27	1.81 (0.63–5.24)	1.88 (0.65–5.44)	
Male	Reference		Reference			Reference	Reference	
Cesarean delivery	1.46 (0.98–2.17)	.06	1.27 (0.84–1.92)	.26	.76	0.86 (0.31–2.36)	1.02 (0.37–2.82)	
Inborn	0.72 (0.45–1.13)	.15	0.96 (0.62–1.49)	.87	.77	1.19 (0.38–3.75)	0.89 (0.28–2.81)	
Psych disorder	0.96 (0.62–1.50)	.85	1.00 (0.65–1.54)	.99	.19	0.45 (0.14–1.49)	0.48, (0.15–1.51)	
MOUD	1.44 (0.76–2.74)	.27	0.88 (0.43–1.79)	.72	.76	1.31 (0.24–7.17)	2.96 (0.47–18.65)	
Any other opioid use	2.66 (1.60–4.40)	<.001	1.16 (0.64–2.09)	.62	.67	0.76 (0.22–2.65)	1.95 (0.49–7.77)	
Any non-opioid substance use ^c	0.88 (0.58–1.35)	.57	1.00 (0.63–1.58)	1.00	.35	1.73 (0.55–5.42)	1.73 (0.57–5.24)	

^a The sources of substance use were derived from maternal report or urine toxicology, or infant urine or meconium toxicology results.

^b The sources of substance use were derived only from toxicology results.

^c Selective serotonin reuptake inhibitor (SSRI) and Gabapentin were included in the model although they were not tested but prescribed.

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NNNS Summary Scores	Clonidine			Morphine			Adjusted <i>P</i> Value Between Treatment		
	Initial	Final	Change	Initial	Final	Change	Initial	Final	Change
Habituation	<i>n</i> = 20, 7.5 (6.0 to 8.3)	<i>n</i> = 23, 8.0 (6.0 to 9.0)	<i>n</i> = 9, 0.5 (−1.0 to 1.0)	<i>n</i> = 33, 7.0 (5.7 to 8.0)	<i>n</i> = 23, 8.0 (7.5 to 8.7)	<i>n</i> = 15, 0.3 (−1.0 to 1.0)	.71	.92	.77
Attention	<i>n</i> = 35, 4.4 (3.3 to 5.3)	<i>n</i> = 48, 5.1 (4.1 to 6.2)	<i>n</i> = 31, 1.0 (−0.1 to 2.0)	<i>n</i> = 42, 4.3 (3.7 to 5.1)	<i>n</i> = 41, 5.3 (4.7 to 5.7)	<i>n</i> = 31, 0.6 (0.2 to 1.6)	.91	.92	.42
Handling	<i>n</i> = 39, 0.6 (0.3 to 0.9)	<i>n</i> = 48, 0.5 (0.4 to 0.8)	<i>n</i> = 35, −0.1 (−0.4 to 0.1)	<i>n</i> = 44, 0.6 (0.4 to 0.8)	<i>n</i> = 46, 0.6 (0.4 to 0.9)	<i>n</i> = 37, 0.1 (0.0 to 0.4)	.91	.44	.01
Quality of movement	<i>n</i> = 56, 4.5 (4.0 to 5.1)	<i>n</i> = 53, 5.0 (4.2 to 5.2)	<i>n</i> = 52, 0.1 (−0.3 to 0.8)	<i>n</i> = 57, 4.8 (4.5 to 5.0)	<i>n</i> = 51, 4.7 (4.3 to 5.2)	<i>n</i> = 51, 0.0 (−0.5 to 0.5)	.47	.61	.42
Regulation	<i>n</i> = 56, 5.1 (4.3 to 5.6)	<i>n</i> = 53, 5.4 (4.9 to 5.9)	<i>n</i> = 52, 0.5 (−0.5 to 1.3)	<i>n</i> = 56, 5.4 (4.9 to 5.8)	<i>n</i> = 51, 5.2 (4.6 to 5.6)	<i>n</i> = 50, −0.2 (−0.8 to 0.2)	.15	.44	.01
Non-optimal reflexes	<i>n</i> = 56, 5.0 (4.0 to 6.0)	<i>n</i> = 54, 4.0 (3.0 to 5.0)	<i>n</i> = 52, −1.0 (−2.5 to 1.0)	<i>n</i> = 57, 5.0 (3.0 to 6.0)	<i>n</i> = 51, 4.0 (2.0 to 4.0)	<i>n</i> = 51, −1.0 (−2.0 to 1.0)	.36	.59	.77
Asymmetry total/reflexes	<i>n</i> = 56, 0.0 (0.0 to 0.0)	<i>n</i> = 54, 0.0 (0.0 to 0.0)	<i>n</i> = 52, 0.0 (0.0 to 0.0)	<i>n</i> = 57, 0.0 (0.0 to 0.0)	<i>n</i> = 51, 0.0 (0.0 to 0.0)	<i>n</i> = 51, 0.0 (0.0 to 0.0)	.99	.69	.51
Arousal	<i>n</i> = 56, 4.0 (3.7 to 4.3)	<i>n</i> = 53, 3.9 (3.4 to 4.1)	<i>n</i> = 52, −0.1 (−0.6 to 0.3)	<i>n</i> = 57, 3.7 (3.3 to 4.1)	<i>n</i> = 51, 3.9 (3.7 to 4.1)	<i>n</i> = 51, 0.4 (0.0 to 0.7)	.04	.44	.004
Hypertonia	<i>n</i> = 56, 1.0 (0.0 to 2.0)	<i>n</i> = 54, 0.0 (0.0 to 0.0)	<i>n</i> = 52, −1.0 (−2.0 to 0.0)	<i>n</i> = 57, 0.0 (0.0 to 1.0)	<i>n</i> = 51, 0.0 (0.0 to 0.0)	<i>n</i> = 51, 0.0 (−1.0 to 0.0)	.04	.92	.06
Hypotonia	<i>n</i> = 56, 0.0 (0.0 to 0.0)	<i>n</i> = 54, 0.0 (0.0 to 1.0)	<i>n</i> = 52, 0.0 (0.0 to 0.5)	<i>n</i> = 57, 0.0 (0.0 to 0.0)	<i>n</i> = 51, 0.0 (0.0 to 0.0)	<i>n</i> = 51, 0.0 (0.0 to 0.0)	.91	.69	.42
Excitability	<i>n</i> = 56, 4.0 (2.0 to 6.0)	<i>n</i> = 53, 3.0 (1.0 to 4.0)	<i>n</i> = 52, −1.0 (−3.0 to 1.0)	<i>n</i> = 57, 3.0 (2.0 to 4.0)	<i>n</i> = 51, 3.0 (1.0 to 5.0)	<i>n</i> = 51, 1.0 (−1.0 to 3.0)	.15	.44	.01
Lethargy	<i>n</i> = 56, 6.0 (4.0 to 8.5)	<i>n</i> = 54, 5.0 (4.0 to 7.0)	<i>n</i> = 52, 0.0 (−3.0 to 1.0)	<i>n</i> = 57, 6.0 (5.0 to 8.0)	<i>n</i> = 51, 4.0 (3.0 to 6.0)	<i>n</i> = 51, −2.0 (−4.0 to 1.0)	.91	.59	.42
Stress abstinence	<i>n</i> = 56, 0.2 (0.1 to 0.2)	<i>n</i> = 53, 0.1 (0.1 to 0.2)	<i>n</i> = 52, 0.0 (−0.1 to 0.0)	<i>n</i> = 57, 0.1 (0.1 to 0.2)	<i>n</i> = 51, 0.1 (0.1 to 0.2)	<i>n</i> = 51, 0.0 (0.0 to 0.1)	.02	.59	.004

* A comparison is shown between treatment groups at the initial administration and at the end of treatment and the change in summary scores from the initial and final assessments. Sample sizes, median, and 25th and 75th percentiles are reported. The false discovery method was used to adjust *P* values for multiple testing. Each item administered is dependent on infant's behavioral state (eg, habituation is elicited during sleep state), and thus, the varying *n* for each item.

of lower mean FS with clonidine treatment probably reflects achieving a steady state. This delayed onset of efficacy for clonidine likely necessitated dose escalations and adjunct medication. We are currently examining our pharmacokinetics data to optimize the dosing regimen with clonidine to shorten the time to steady state.

The worse neurobehavioral performance of the clonidine-treated infants compared to the morphine group on the initial assessment may be explained by the delayed or lesser effectiveness of the clonidine response or not having a designated specific time point for the initial assessment. The improvement in neurobehavioral performance with clonidine treatment is reassuring; neurobehavioral functioning is reported to predict early childhood outcomes in drug-exposed infants.⁵¹ Long-term follow-up is needed to understand the implications of Nows and its various treatment modalities.

This study was not powered to show equivalence or non-inferiority. Having this limitation, the lack of significant difference may reflect a type II hypothesis testing error, which would become less probable with a larger sample size. The coronavirus disease 2019 pandemic impacted our plans to enroll a larger sample and add

another study site. Nonetheless, the number of infants randomly assigned in our study was similar to that of other multicenter treatment trials.^{58,79} Being a single study site limits the generalizability of findings. However, we provide information on infants from rural areas. Some mothers did not room in with their babies, and many did not initiate breastfeeding; these affect non-pharmacological care and LOS.^{80,81} Although a high proportion of mothers were on MOUD, the use of other substances was common. Polypharmacy may also influence Nows treatment.⁸² Ascertaining the effects of polydrug use will require a larger number of participants.

In summary, in a randomized trial, the length of pharmacologic treatment and the final assessment of neurobehavioral performance were not significantly different between the clonidine- and morphine-treated groups. Future studies are needed to investigate the optimal dose and frequency of clonidine administration for improved efficacy and the decreased need for adjunctive therapy in Nows.

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ABBREVIATIONS

CI: confidence interval
CONSORT: Consolidated Standards of Reporting Trials
DSMB: Drug Safety Monitoring Board
FS: Finnegan Neonatal Abstinence Scoring System score
HR: hazard ratio
IRB: institutional review board
LOS: length of stay
LOT: length of treatment
MOUD: medication for opioid use disorder
NAS: neonatal abstinence syndrome
NOWS: neonatal opioid withdrawal syndrome
NNNS: NICU network neurobehavioral scale

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